

UST-KUT, Russian Federation

WMO: 303200

Lat: 56.857N

Lon: 105.730E

Elev: 668

StdP: 93.55

Time Zone: 8.00 (E08)

Period: 90-00

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -40.0	(c) -36.4	(d) -41.7	(e) 0.1	(f) -39.6	(g) -38.1	(h) 0.1	(i) -36.0	(j) 9.3	(k) -12.8	(l) 8.4	(m) -15.2	(n) 0.0	(o) 270	(p) 0.681

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	28.2	17.4	26.3	17.1	24.6	16.2	19.3	25.3	18.4	24.2	17.4	22.8	2.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	13.5	22.1	16.3	12.6	20.9	15.3	11.8	20.0	58.2	25.1	55.1	24.5	51.9	23.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
8.5	7.4	6.5		-37.9	31.2	3.3	2.4	-40.3	32.9	-42.2	34.3	-44.0	35.7	-46.4	37.5	
				DB				-40.3	32.9	-42.2	34.3	-44.0	35.7	-46.4	37.5	
				WB	-38.0	21.5	3.2	1.6	-40.3	22.6	-42.2	23.5	-44.0	24.4	-46.4	25.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.9	-22.6	-15.7	-9.7	-1.8	7.3	13.7	18.3	15.2	6.0	-1.6	-14.0	-19.0		
	DBStd	14.85	8.98	5.61	6.98	4.06	5.51	4.37	3.81	3.71	3.90	5.53	8.54	7.46		
	HDD10.0	4942	1012	719	609	353	124	13	1	4	130	359	720	898		
	HDD18.3	7459	1270	953	868	603	344	151	49	108	370	617	970	1156		
	CDD10.0	597	0	0	0	0	41	123	257	166	11	0	0	0		
	CDD18.3	71	0	0	0	0	2	11	47	11	0	0	0	0		
	CDH23.3	647	0	0	0	0	44	121	383	99	0	0	0	0		
	CDH26.7	132	0	0	0	0	9	24	86	13	0	0	0	0		
(14) Wind	WSAvg	2.9	2.6	2.8	3.0	3.7	3.3	2.7	2.2	2.3	2.7	3.2	3.3	3.2		
(15) Precipitation	PrecAvg	488	25	20	18	24	35	58	73	79	48	41	38	30		
	PrecMax	586	42	36	44	49	64	111	146	142	83	71	56	63		
	PrecMin	360	7	9	3	10	9	19	26	46	5	12	23	15		
	PrecStd	57	9	7	9	9	13	21	31	27	17	14	9	11		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.9	-0.4	7.9	12.3	27.5	29.7	31.2	28.3	19.4	13.1	3.0	-2.3		
		MCWB	-4.8	-3.2	1.8	4.5	14.4	17.0	19.1	17.6	12.8	6.6	0.9	-3.4		
	2%	DB	-6.1	-3.8	4.4	9.1	24.0	26.6	29.1	25.9	15.9	10.2	1.8	-4.3		
		MCWB	-6.4	-5.1	0.2	3.1	12.7	16.2	18.1	17.6	10.5	5.0	0.4	-5.3		
	5%	DB	-8.7	-6.2	2.0	6.3	20.4	24.3	27.2	23.9	13.7	7.7	-0.7	-7.1		
		MCWB	-9.3	-7.3	-1.5	1.1	11.3	15.3	17.6	16.2	9.3	4.1	-1.8	-7.7		
	10%	DB	-11.0	-8.4	-0.3	4.3	17.1	22.0	25.5	21.8	11.9	5.5	-3.3	-9.3		
		MCWB	-11.4	-9.2	-2.9	0.3	9.8	14.8	17.3	15.6	8.4	2.6	-4.2	-9.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.5	-2.8	2.2	5.0	15.7	19.7	20.7	20.1	13.1	7.2	1.4	-3.4		
		MCDB	-4.0	-1.2	6.4	10.3	25.1	25.5	26.6	24.9	17.6	10.6	2.5	-2.5		
	2%	WB	-6.5	-5.0	0.5	3.8	13.6	17.8	19.7	18.5	11.4	5.7	0.1	-5.3		
		MCDB	-6.1	-4.0	4.0	8.2	21.4	24.2	25.7	23.3	15.1	9.4	1.6	-4.5		
	5%	WB	-9.3	-7.2	-1.1	2.1	11.9	16.4	19.0	17.4	10.0	4.4	-1.7	-7.6		
		MCDB	-8.7	-6.3	1.5	5.2	19.6	22.0	25.3	22.2	12.7	7.6	-0.8	-7.1		
	10%	WB	-11.5	-9.2	-2.7	0.6	10.1	15.1	18.2	16.3	9.1	2.7	-4.0	-9.8		
		MCDB	-11.1	-8.4	-0.2	3.5	16.5	21.2	23.9	21.0	11.2	5.0	-3.3	-9.3		
(35) Mean Daily Temperature Range	MDBR		6.2	8.3	10.4	8.2	10.7	10.4	10.1	9.5	7.3	7.0	6.6	6.7		
	5% DB	MCDBR	6.5	10.1	12.1	11.5	16.9	14.5	13.7	13.5	10.1	10.2	7.9	7.8		
		MCWBR	6.0	8.7	8.8	7.4	9.0	7.4	5.7	6.4	5.5	6.8	7.0	7.6		
	5% WB	MCDBR	6.2	9.3	11.8	9.7	15.4	11.8	10.8	11.7	8.9	9.3	7.9	8.0		
		MCWBR	5.8	8.1	8.9	6.7	8.4	7.0	5.3	6.4	5.3	6.7	7.1	7.8		
(40) Clear-Sky Solar Irradiance	taub		0.261	0.291	0.313	0.342	0.384	0.413	0.444	0.387	0.329	0.311	0.273	0.249		
	taud		2.020	2.035	2.073	2.167	2.253	2.251	2.172	2.325	2.464	2.337	2.086	1.981		
	Ebn at Noon		590	737	831	868	852	828	789	817	827	729	571	485		
	Edh at Noon		61	93	117	126	124	126	134	108	81	69	57	48		
(44) All-Sky Solar Radiation	RadAvg		0.75	1.71	3.10	4.44	5.18	5.71	5.27	4.30	2.73	1.41	0.74	0.47		
	RadStd		0.07	0.12	0.17	0.31	0.38	0.45	0.51	0.29	0.26	0.16	0.09	0.07		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page